Secular coefficients mini-course
Talk 2

Ofir Gorodetsky, Technion

ofir-gorodetsky.github.io/Bristol

CMI-HIMR Summer School, Bristol
June 2, 2026

Recall the analogy:

$$L(s, \chi) \text{ (for random } \chi) \text{ or } \zeta(s + it) \text{ (random } t)$$

are analogous to

$$\det(I - uU_N),$$

the characteristic polynomial of a random unitary matrix U_N chosen according to Haar measure.

The coefficients of $\det(I - uU_N)$ are called *secular coefficients*:

$$\det(I - uU_N) = \sum_{i=0}^{N} (-1)^i \mathrm{Sc}_i(U_N) u^i.$$

What is their distribution?

From the identity

$$\det(I - uU_N) = \exp\left(-\sum_{n=1}^{\infty} \frac{\mathrm{Tr}(U_N^n)u^n}{n}\right)$$

it follows that $\mathrm{Sc}_i(U_N)$ is a polynomial of total degree i in $(\mathrm{Tr}(U_N^n))_{n\leq i}$.

Since the tuple $(\mathrm{Tr}(U_N^n))_{n\leq i}$ tends to a tuple of independent Gaussians, $\mathrm{Sc}_i(U_N)$ tends to a degree-i polynomial in independent Gaussians.

Harder to study when i is growing.

Is there a number-theoretic counterpart for $\mathrm{Sc}_i(U_N)$? Yes:

$$\frac{1}{\sqrt{x}}\sum_{n\leq x}\chi(n) \text{ or } \frac{1}{\sqrt{x}}\sum_{n\leq x}n^{it},$$

for random χ or t. These are important and well-studied objects in number theory ('character sums'), which implies secular coefficients are worth studying.

For instance: the Lindelöf hypothesis states that $\zeta(1/2+it)\ll_\varepsilon (1+|t|)^\varepsilon$. Equivalent to

$$\sum_{n\leq x}n^{it}=\int_1^x u^{it}du+O_\varepsilon(\sqrt{x}(x(1+|t|))^\varepsilon).$$

The analogy between character sums and secular coefficients is more convincing if one considers polynomials over finite fields: recall

$$\mathcal{L}(u,\chi) = \sum_{f \text{ monic}} \chi(f) u^{\deg f}$$

is a polynomial, and $\mathcal{L}(u/\sqrt{q},\chi) = \det(I - u\Theta_\chi)$ for some unitary matrix (that equidistributes as χ varies, by Katz).

The ith coefficient of $\mathcal{L}(u/\sqrt{q},\chi)$ is precisely a normalized character sum:

$$q^{-i/2} \sum_{\deg f = i} \chi(f).$$

For fixed i, this is a polynomial in $(\sum_{\deg f=n} \chi(f)\Lambda(f))_{n\le i}$ which tends to a tuple of independent random variables (when χ is sampled uniformly modulo M with $\deg M \to \infty$).

Recall $\det(I - uU_N)$ tends to

$$F(u) := \exp\left(-\sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}}\right),$$

where $(N_i)_i$ are iid standard complex Gaussians.

Studying coefficients of $F(u)$ should be easier than studying secular coefficients. We write

$$F(u) = \sum_{i=0}^{\infty} c_i u^i.$$

We want to compute $\mathbb{E}[|c_n|^2]$. By independence of the N_i,

$$\mathbb{E}[F(u)\overline{F}(z)] = \mathbb{E}\exp\left(-\sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}} - \overline{N_i}\frac{z^i}{\sqrt{i}}\right)$$

$$= \prod_{i=1}^{\infty} \mathbb{E}\exp\left(-N_i \frac{u^i}{\sqrt{i}} - \overline{N_i}\frac{z^i}{\sqrt{i}}\right) \qquad = \prod_{i=1}^{\infty} \exp\left(\frac{z^i u^i}{i}\right)$$

$$= \exp\left(-\log(1-zu)\right)$$

$$= \frac{1}{1-zu} = 1 + zu + (zu)^2 + \dots$$

Comparing coefficients:

$$\mathbb{E}|c_n|^2 = \mathbb{E}[c_n\overline{c_n}] = 1.$$

More generally, $\mathbb{E}[c_n\overline{c_m}] = \delta_{n,m}$.

Similarly, instead of studying $\frac{1}{\sqrt{x}}\sum_{n\leq x}\chi(n)$ for random χ we may want to begin with the easier question of studying

$$\frac{1}{\sqrt{x}}\sum_{n\leq x}\alpha(n)$$

where α is the Steinhaus random multiplicative function: $(\alpha(p))_p$ are iid.

We have perfect orthogonality for α (exercise):

$$\mathbb{E}[\alpha(n)\overline{\alpha}(m)] = \delta_{n,m}.$$

This implies

$$\mathbb{E}|\sum_{n\le x}\alpha(n)|^2 = \lfloor x \rfloor,$$

so by Cauchy–Schwarz, $\mathbb{E}|\sum_{n\le x}\alpha(n)| \le \sqrt{x}$. A celebrated result of Harper says that the correct size of the first moment is smaller than that.

Theorem 1 (Harper, 2017)

We have

$$\frac{c\sqrt{x}}{(\log\log x)^{1/4}} \le \mathbb{E}|\sum_{n\le x}\alpha(n)| \le \frac{C\sqrt{x}}{(\log\log x)^{1/4}}.$$

More recently, Harper has deduced a ‘deterministic’ consequence.

Theorem 2 (Harper, 2023)

Let m be a prime. Uniformly for $m \geq x \geq 1$,

$$\frac{1}{m-1} \sum_{\chi \bmod m} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \chi(n) \right| \leq \frac{C}{(\log\log \min\{x, m/x\})^{1/4}}.$$

Uniformly for $T \geq x \geq 1$,

$$\frac{1}{T} \int_0^T \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} n^{it} \right| dt \leq \frac{C}{(\log\log \min\{x, T/x\})^{1/4}}.$$

For c_i, Najnudel, Paquette and Simm proved the following results.

Theorem 3 (Najnudel–Paquette–Simm, 2020)

We have

$$\mathbb{E}\,|c_n| \leq \frac{C}{(\log n)^{1/4}}.$$

Theorem 4 (Najnudel–Paquette–Simm, 2020)

For $0 \leq n \leq N$,

$$\mathbb{E}\,|\mathrm{Sc}_n(U_N)| \leq \frac{C}{(\log \min\{n, N-n\})^{1/4}}.$$

Exercise: $|\mathrm{Sc}_n(U_N)|$ and $|\mathrm{Sc}_{N-n}(U_N)|$ have the same distribution.

The proof of Najnudel–Paquette–Simm shares similarities to Harper's, but there are differences as well.

Soundararajan and Zaman (2021) proved $\frac{c}{(\log n)^{1/4}} \leq \mathbb{E}\,|c_n| \leq \frac{C}{(\log n)^{1/4}}$, with a proof mirroring Harper closely.

Recently there was progress on the limiting distribution of

$$\frac{1}{\sqrt{x}(\log\log x)^{-1/4}} \sum_{n \le x} \alpha(n)$$

by Gorodetsky–Wong, and on the limiting distribution of

$$(\log n)^{1/4} c_n$$

by Atherfold–Najnudel. More on that in the next lecture.

The simplest approach to proving that a sequence of random variables X_n tends in distribution to X is the method of moments. For real random variables it says: if

$$\lim_{n\to\infty} \mathbb{E}[X_n^k] = \mathbb{E}[X^k]$$

holds for every positive integer k, and if $\sum_{k\geq 1} \mathbb{E}[X^{2k}]^{-1/(2k)}$ diverges, then $X_n \xrightarrow{d} X$.

Example: if $X \sim N(0,1)$ then $\mathbb{E}[X^{2k}]^{-1/(2k)} \sim c/\sqrt{k}$.

For both $\sum_{n\leq x} \alpha(n)$ and c_n, even after normalization, the moments diverge (see exercises), so this approach cannot work. Indicates the distribution has heavy tails.

Yet, various central limit theorems are known! Given a subset $\mathcal{A}_x \subseteq \{1, 2, \ldots, x\}$, consider

$$S(x, \mathcal{A}_x) := |\mathcal{A}_x|^{-1/2} \sum_{n \in \mathcal{A}_x} \alpha(n).$$

In some cases, it is known that $S(x, \mathcal{A}_x) \xrightarrow{d} N_{\mathbb{C}}(0, 1)$:

1. $\mathcal{A}_x$ is the short interval $[x - h, x]$ where $h \leq x/(\log x)^c$ for suitable $c > 0$ (Chatterjee–Soundararajan, Soundararajan–Xu, Pandey–Wang–Xu)
2. $\mathcal{A}_x$ consists of integers with k prime factors, $k = o(\log \log x)$ (Hough, Harper)
3. $\mathcal{A}_x$ consists of elements of $Q(\mathbb{N})$ (Najnudel, Klurman–Shkredov–Xu, Chinis–Shala)

All the known examples are sparse: $|\mathcal{A}_x| = o(x)$.

What are the proof methods?

1. Method of moments
2. McLeish CLT

Method of moments

Let $X \sim N_{\mathbb{C}}(0,1)$. To prove that $S(x,\mathcal{A}_x) \xrightarrow{d} X$, it suffices to prove that

$$\lim_{x\to\infty} \mathbb{E}[S(x,\mathcal{A}_x)^{k_1}\overline{S(x,\mathcal{A}_x)}^{k_2}] = \mathbb{E}[X^{k_1}\overline{X}^{k_2}] = k_1!\delta_{k_1,k_2}.$$

For $k_1 = k_2 = k$, this becomes

$$\#\left\{ {}^{n_1,\ldots,n_k\in\mathcal{A}_x}_{m_1,\ldots,m_k\in\mathcal{A}_x},\ n_1\cdots n_k = m_1\cdots m_k \right\} \sim k!|\mathcal{A}_x|^k.$$

Given a permutation $\pi \in S_k$, the terms with $n_i = m_{\pi(i)} \in \mathcal{A}_x$ contribute $|\mathcal{A}_x|^k$, so the task becomes bounding the *off-diagonal contribution*: terms where one cannot find a permutation π such that $n_i = m_{\pi(i)}$ for all i.

It turns out that for the random variable $S(x, \mathcal{A}_x)$, there is a useful *fourth moment criterion*.

Theorem 5 (Soundararajan–Xu)

Let $\mathcal{A}_x \subseteq \{1, 2, \dots, x\}$. Suppose the following two conditions hold:

- $\lim_{x\to\infty} \mathbb{E}|S(x, \mathcal{A}_x)|^4 = 2$.
- *Let $P(n)$ be the largest prime factor of n. For every prime $p \leq x$, we have $|\{n \in \mathcal{A}_x : P(n) = p\}| = o(|\mathcal{A}_x|)$.*

Then

$$S(x, \mathcal{A}_x) \xrightarrow{d} N_{\mathbb{C}}(0, 1).$$

Adapted to sums of the Steinhaus function in $\mathbb{F}_q[T]$ (Hoban–Shah–Ismail–Verreault–Zaman).

What does the fourth moment condition actually say? That

$$\#\left\{(a,b,c,d) \in (\mathcal{A}_x)^4 : ab = cd, \, {\scriptstyle (a,b)\neq(c,d) \atop (a,b)\neq(d,c)}\right\} = o(|\mathcal{A}_x|^2).$$

One can squeeze out a bit more. Let $\mathcal{B}_x \subseteq \mathcal{A}_x$. If $|\mathcal{B}_x| = o(|\mathcal{A}_x|)$ then

$$\frac{\sum_{n\in\mathcal{A}_x}\alpha(n)}{\sqrt{|\mathcal{A}_x|}} \xrightarrow{d} N_{\mathbb{C}}(0,1) \longleftrightarrow \frac{\sum_{n\in\mathcal{A}_x\setminus\mathcal{B}_x}\alpha(n)}{\sqrt{|\mathcal{A}_x|}} \xrightarrow{d} N_{\mathbb{C}}(0,1).$$

The reason: considering the difference, its second moment vanishes in the limit, hence it tends in distribution to 0.

Consequence: We may apply the fourth moment *after* discarding a sparse set.

We wish to explain what is the McLeish Central Limit Theorem and how it gives the criterion. Relies on martingales.

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.
Let $(\mathcal{F}_i)_{i \geq 1}$ be a filtration: an increasing sequence of sub-σ-algebras of $\mathcal{F}$.
A sequence of random variables $(R_i)_{i \geq 1}$ is called a *martingale adapted to* $(\mathcal{F}_i)_{i \geq 1}$ if R_i is measurable with respect to $\mathcal{F}_i$ for all $i \geq 1$, and

$$\mathbb{E}[R_i \mid \mathcal{F}_{i-1}] = R_{i-1}$$

for all $i \geq 2$.

Theorem 6 (McLeish, 70s)

For any $n \geq 1$, let $(X_i^{(n)})_{i\geq 1}$ be a martingale difference sequence, i.e. $(X_1^{(n)}+\dots+X_i^{(n)})_{i\geq 1}$ is a martingale. A sufficient condition for

$$\sum_{i\geq 1} X_i^{(n)} \xrightarrow{d} N_{\mathbb{C}}(0,1)$$

to hold as $n\to\infty$ is

$$\lim_{n\to\infty} \sum_{i\geq 1} \mathbb{E}[|X_i^{(n)}|^4] = 0, \qquad \lim_{n\to\infty} \mathbb{E}[|\sum_{i\geq 1}(X_i^{(n)})^2|^2] = 0,$$

$$\lim_{n\to\infty} \mathbb{E}\left[\Big(\sum_{i\geq 1}|X_i^{(n)}|^2-1\Big)^2\right] = 0.$$

Exercise: if $(X_i^{(n)})_i$ are independent with $\mathbb{E}|\sum_i X_i^{(n)}|^2 = 1$ then 1st condition implies the 3rd and we recover (complex) CLT.

We now relate the McLeish CLT to $S(x, \mathcal{A}_x)$. If $P(n)$ is the largest prime factor of n then we may write (Harper)

$$\sum_{n \in \mathcal{A}_x} \alpha(n) = \sum_{p \leq x} X_p$$

where

$$X_p = \sum_{n \in \mathcal{A}_x \colon P(n) = p} \alpha(n).$$

The sequence $(\sum_{p \leq q} X_p)_{q \leq x}$ is a martingale with respect to the filtration $\mathcal{F}_q = \sigma(\alpha(p) : p \leq q)$ indexed by primes.

Suffices to check: if q is the largest prime less than q then $\mathbb{E}[X_p \mid \alpha(2), \alpha(3), \ldots, \alpha(q)] = 0$. Reduces to $\mathbb{E}[\alpha(p)^k] = 0$ for $k \geq 1$.

Recall $(c_n)_{n\geq 0}$ are defined via

$$\sum_{n=0}^{\infty} c_n u^n = \exp\left(-\sum_{i=1}^{\infty} \frac{N_i u^i}{\sqrt{i}}\right).$$

Given n, we write $\lambda \vdash n$ to indicate that λ is a partition of n, and write $a_i(\lambda)$ for the number of parts of size i in λ.

Expanding the definition of c_n we see that there are coefficients $(b_\lambda)_{\lambda\vdash n}$ such that

$$c_n = \sum_{\lambda\vdash n} b_\lambda \prod_{i=1}^{n} N_i^{a_i(\lambda)}.$$

Given an expression

$$A(n,\mathbf{b}) := \sum_{\lambda \vdash n} b_\lambda \prod_{i=1}^{n} N_i^{a_i(\lambda)},$$

we decompose it as

$$A(n,\mathbf{b}) = \sum_{i=1}^{n} Z_i$$

where Z_i consists of terms in $A(n,\mathbf{b})$ corresponding to partitions λ whose largest part is of size i.

The expression Z_i involves a power of N_i but does not involve N_j for $j > i$. The sequence $(\sum_{i=1}^{m} Z_i)_{1 \le m \le n}$ is a martingale with respect to the filtration $\mathcal{F}_m = \sigma(Z_1, \ldots, Z_m)$ $(1 \le m \le n)$.

Thank you for listening.